

Project Planning — Selin Tuerkoglu

Goals (what you are hoping to achieve)

Define your main goal in a single sentence (can be somewhat broad). Explain how you understand this goal (a few sentences) and introduce a limited number of sub-goals. Sub-goals should be clear and actionable, directly relevant to the main goal and defined with consideration to the practical constraints (available datasets, analysis methods, time, etc.). These can be revised and adjusted later if necessary.

Main Goal:

To develop a domain-specific pan-cancer single-cell foundation model capable of predicting pharmacological response (efficacy/toxicity), which can subsequently be fine-tuned for specific cancer types and/or clinical datasets (binary clinical outcomes).

Because the exact architectural setup depends heavily on label viability, there are two potential pathways to define the initial training target:

- **Goal Option A (Single-Task Efficacy Focus):** To predict continuous pharmacological response (viability/IC50) by mapping single-cell RNA-seq profiles to a curated set of compounds from a single high-coverage database.
- **Goal Option B (Multi-Task Broad Focus):** To simultaneously predict multiple response metrics (efficacy and toxicity) across pan-cancer cell lines using a multi-task learning setup (via masked losses) to handle sparse or missing labels.

Understanding of the Goal:

By relying on continuous experimental labels, we can establish a data-driven model linking cellular heterogeneity to drug response. Since continuous labels capture a more complex distribution of response than binary cutoffs, pre-training on these will create a richer representation that can later be fine-tuned for clinical datasets with binary outputs. However, a primary challenge is a resolution mismatch: standard pharmacological labels (CTRPv2_[1], PRISM_[2], GDSC_[3]) are derived from bulk cell populations, whereas our transcriptomic data (SCP542_[4]) captures individual cellular

heterogeneity. Our overarching goal must therefore include strategies to bridge this bulk-to-single-cell gap.

Sub-goals:

1. **Compound Harmonization:** Finalize a unified drug catalog by matching compound identities across GDSC, CTRPv2, and PRISM using normalized names and BRD-based canonical IDs, while mapping subsets to DrugBank_[5] for future FDA status filtering.
2. **Managing Matrix Sparsity:** Design a training approach to accommodate incomplete bulk-derived label matrices across different databases.
3. **Baseline Prototyping:** Establish the initial predictive baseline using the highest-confidence data intersection (190 overlapping cell lines and 545 compounds between SCP542 and CTRPv2 (Fig. 1)) before attempting complex dataset integration.
4. **Stretch Goal (Time Permitting) - Feature Interpretability:** If the baseline regression model is successfully established and time allows, we will employ Explainable AI (XAI) methods to extract feature importance. This will allow us to bridge the predictive model back to underlying biological mechanisms by highlighting the key transcriptomic drivers of drug resistance.

	Dataset	Total Cell Lines	Overlap with SCP542	Missing Lines	Target Metric	Drug Count
0	GDSC	967	133	65	LN_IC50 / AUC	295
1	CTRPv2	1107	190	8	Viability / AUC	545
2	PRISM	915	182	16	Viability	6575

(Fig. 1 showing the coverage of the 3 different annotation datasets GDSC, CTRP, PRISM compared to the scRNA-seq data set SCP542)

	dataset	target_metric	non_null_rows_own_cell_lines	rows_in_overlap_subset	non_null_rows_overlap_scp542	pct_overlap_vs_own	pct_non_null_within_overlap_subset
0	GDSC	LN_IC50	242036	34738	34738	14.352410	100.000000
1	CTRPv2	cpd_avg_pv (viability)	7227951	1521028	1521028	21.043696	100.000000
2	PRISM	extended primary matrix value (viability proxy)	4213048	1235780	1210432	28.730553	97.948826

(Fig. 2 showing the applicable sample numbers (non-null response values) for SCP542-overlapping cell lines)

Value and strategy (why and how you are doing it)

Explain why what you are doing is valuable (few sentences) – this will help to determine the overall direction of the work. For example, if the value is in “improving patient treatment ...” vs. “discovery of important biological mechanisms ...”, different types of data, analysis and success criteria will be employed. Then, clarify in which way what you are doing is new and original and was not already done in previous research. Outline your strategy for achieving this: what assumptions are you making and carefully examine why you actually believe your chosen approach will be effective? (e.g.: a new technique / model that was successful in another domain can be applied because the data structure is similar).

Doing this part requires reviewing some papers, so it is preferable to also organize these for potential future use as references to save time later, like starting a reference manager library with some comments on why you thought the paper was relevant.

Value:

Understanding how tumor heterogeneity contributes to drug resistance and treatment failure is critical, yet this resolution is completely obscured when relying on traditional bulk RNA-seq models. While predicting drug response directly at the single-cell level holds the key to this challenge, the immediate bottleneck is the lack of large-scale clinical data for model training. Therefore, the value of this project lies in establishing the necessary computational foundation.

Because we lack single-cell-resolution drug screening data at scale, we will employ a weakly supervised approach by mapping bulk pharmacological labels to individual cells of the corresponding *in vitro* cell lines. While laboratory cell lines do not fully recapitulate clinical tumor environments, they provide the high-volume, continuous data required to train robust foundation models. By forcing the model to map heterogeneous single-cell inputs to an average bulk response, it learns the transcriptomic signatures of sensitivity. Post-training, this allows us to perform inference on individual cells and predict a distribution of sensitivities within a single sample. This enables the computational identification of rare, naturally resistant sub-clones before selection pressure is ever applied, directly addressing how baseline tumor heterogeneity drives resistance. Ultimately, this creates a highly generalizable pre-trained model that can subsequently be fine-tuned on scarce clinical data to improve patient-specific treatment predictions.

Originality:

While the original PERCEPTION_[6] study established a baseline by pre-training on bulk data and predicting PRISM AUC labels, this project diverges by focusing on modern representation learning to create a reusable foundation model. Specifically, rather than relying solely on custom bulk-to-single-cell pre-training, this project aims to evaluate multi-task learning across disparate databases (CTRPv2, PRISM, GDSC) and leverage state-of-the-art transformer embeddings (scGPT_[7]) to handle single-cell sparsity and noise.

Strategy:

The core strategy is to use single-cell data (SCP542) to capture cellular heterogeneity and combine this with pharmacological labels derived from bulk assays. Because raw scRNA-seq data is notoriously sparse and noisy, directly mapping it to bulk response labels is highly unstable. To solve this, the strategy relies on utilizing foundation model embeddings (e.g., scGPT) to bridge the bulk-to-single-cell gap.

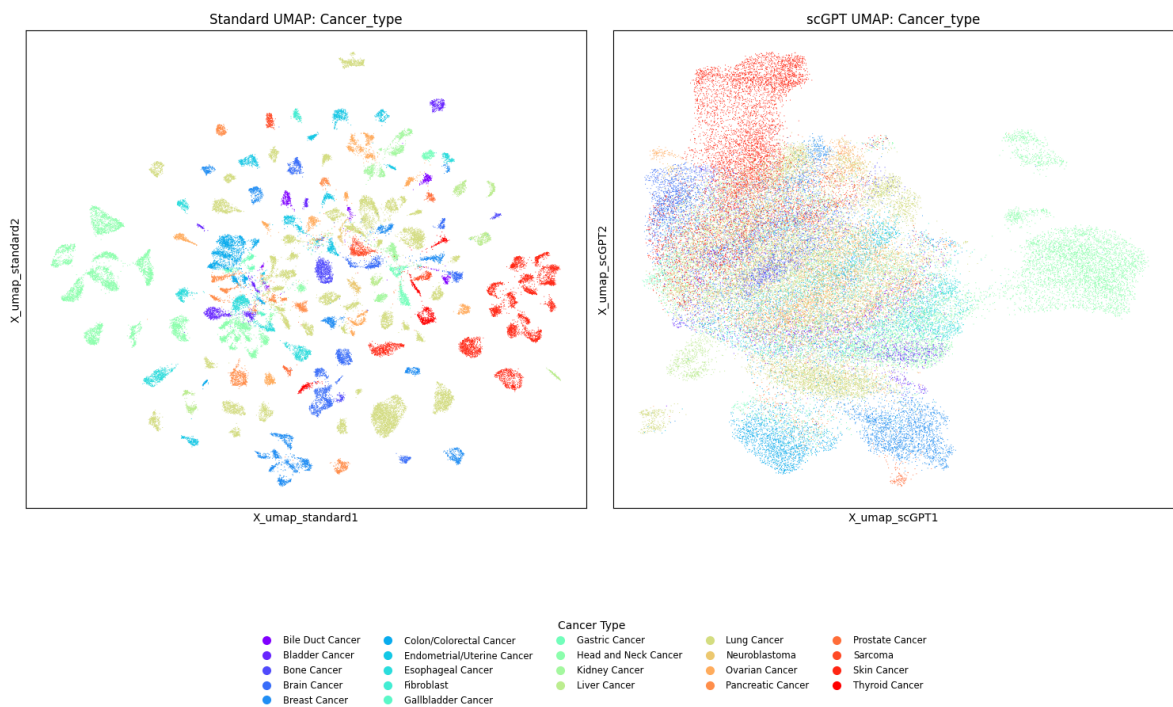
A critical challenge in this weakly supervised approach is that mapping an average bulk drug score onto highly heterogeneous single cells introduces significant label noise. Initial exploratory UMAP analysis confirms this phenomenon and highlights the limitations of standard linear feature extraction. When colored by cancer type (Fig. 3), standard full-transcriptome Principal Component Analysis (PCA) heavily biases towards dominant tissue-specific markers, artificially clustering cells into distinct lineage "islands." In contrast, scGPT embeddings successfully overcome this bias, projecting cells into a continuous, shared pan-cancer manifold.

When evaluating a specific drug example mapped from CTRPv2 (Paclitaxel), the impact of this difference becomes critical (Fig. 4). Because standard PCA isolates cell lines, it merely memorizes the bulk labels, painting each island a uniform color. If used for training, a downstream model would "cheat" by simply classifying the cell line rather than learning drug resistance. Conversely, the scGPT manifold displays a highly mixed distribution of sensitivity. This visually confirms that while foundation models cannot isolate pharmacological sensitivity unsupervised, they successfully strip away the artificial tissue bias, leaving a true, unified representation of cellular states.

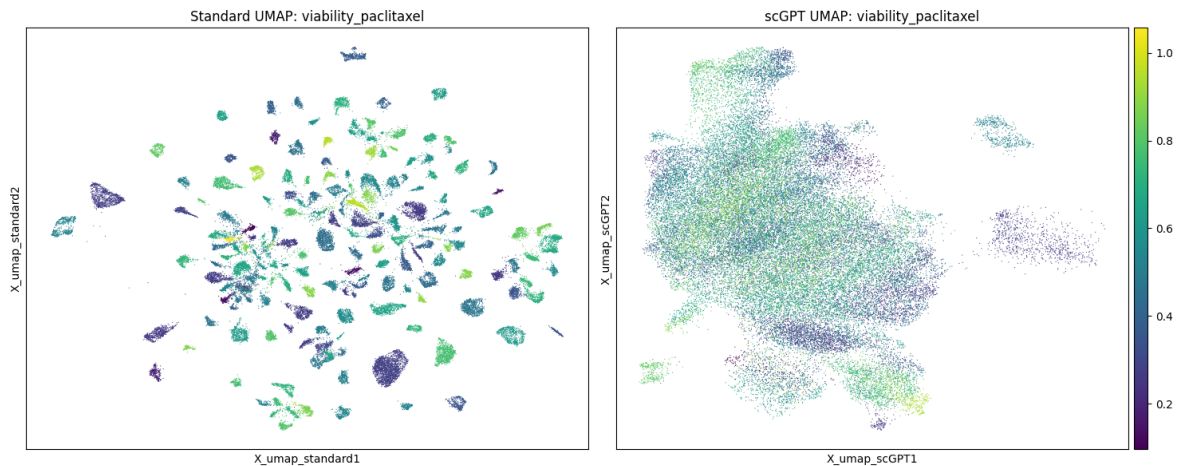
Core Hypothesis: Our central hypothesis is that scGPT embeddings provide a superior, denoised biological prior precisely because they force the alignment of

functional cellular states across disparate tissues. We hypothesize that by training a supervised regression architecture on this specific manifold, the model will be forced to learn genuine transcriptomic drivers of resistance rather than merely classifying tissue-of-origin.

To validate this architectural choice, our exploratory UMAP visualizations (Fig. 3 and Fig. 4) serve as the initial proof of concept. They visually confirm that the foundation model acts as a successful prior, overcoming the tissue-specific memorization inherent to standard PCA and providing a rigorous, generalized starting point for the downstream supervised regression task.



(Fig. 3: Comparative UMAP analysis colored by Cancer Type. The standard full-transcriptome PCA baseline (left) artificially forces cells into distinct, lineage-specific clusters based on tissue of origin. The scGPT embeddings (right) overcome this bias, projecting the pan-cancer dataset into a continuous, shared functional manifold.)



(Fig. 4: Comparative UMAP analysis colored by mapped CTRPv2 Paclitaxel viability. The standard PCA baseline (left) merely memorizes cell lines, resulting in uniformly colored islands. Due to the weakly supervised mapping of average bulk scores onto highly heterogeneous single cells, the visual distribution of sensitivity across the scGPT manifold (right) appears mixed, confirming the necessity of a supervised regression architecture to extract pharmacological signals.)

Prototyping and implementation

Wherever possible, pre-processed datasets and established protocols from previous papers should be used when preparing the data. It is acceptable to use publicly shared code for this part and evaluation of results, when available. The model specification and training code should be original and project-specific, simply extending some existing core code is only acceptable for your own previous work or that of your current lab (with permission of original authors).

The starting point should always be small and manageable – addressing only part of the planned requirements and only some of all available data is ideal for the initial stage. Alternatively, first simply replicating previously published results with a newly developed model is also fine. The model should be as small as possible while still being functional and capturing key elements of the chosen architecture. Starting with a neural network that is too large will waste a lot of time due to issues with stability and difficulties with parameter optimization.

Planned new functionality should be added step-by-step, by creating new versions and aiming to test relevant assumptions identified during the planning stage where

relevant. These different working versions, data necessary to re-run them and results produced should be documented and retained in some way - even for cases where the results were deemed suboptimal.

It is important to keep in mind that things may not always workout in a favorable way, and therefore making detailed plans beyond what was already confirmed to work will likely be counterproductive. Having full clarity about the hypotheses or assumptions being made is therefore critical for figuring out when a strategy should be considered unsuccessful and a re-think is needed. The key is to identify such failures as quickly as possible and using the least resource-expensive ways (e.g.: literature, exploratory analysis, simple statistical tests). Even in such cases, there will usually still be code that can be re-purposed and something new learned about problem to recover easily and continue.

To ensure stability and rapid iteration, the initial prototype will be strictly constrained. Rather than attempting to build the full multi-task architecture immediately, the project will begin with a simplified baseline that addresses only a portion of the available data and focuses on a single predictive task.

Data: I will begin exclusively with the highest-confidence dataset intersection: SCP542 and CTRPv2. My exploratory data audit (Fig. 1) confirmed an overlap of 190 unique cell lines and 545 compounds, with 100% non-null viability scores within that specific overlap subset (Fig. 2), providing a complete target matrix for initial training.

Model: The first phase will focus entirely on latent space validation, generating the scGPT embeddings and conducting the comparative UMAP analysis against standard PCA features to verify the structural integrity of the biological prior. Once visually validated, the first predictive iteration will be a simple supervised regression architecture aiming to predict the continuous cpd_avg_pv (viability) score using these embedded representations.

Implementation: I will not attempt the multi-task setup or integrate PRISM and GDSC data until this simplified single-task baseline is functioning. Once the embedding pipeline is established and the baseline regression model is successfully optimized on CTRPv2, I will iteratively expand the architecture to include masked losses and incorporate the much larger, highly sparse PRISM dataset.

Once the embedding pipeline is established and the full architecture is stable, any remaining project time will be dedicated to applying Explainable AI (XAI) techniques.

This late-stage analysis will help satisfy the biological discovery aspect by interpreting the learned feature interactions driving the model's sensitivity predictions.

References

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